

Preferences and Utility

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How to start modeling choices?

- Last lecture we analyzed the consumer's budget set $B(p, m)$.
- So how to start modeling how individuals make choices?
- We will describe an economic agent only by his attitude towards the bundles in a relevant choice set, i.e. $x \in X \subset \mathbb{R}_+^n$
- We will assume that his attitude is expressed in the form of **preferences**.
- Today, we seek to come up to with a “proper” way to formalize the concept of preferences.
- After studying this concept, we will be concerned with when it is possible to represent preferences by “something handy”, called an **utility function**.

- Although the goal is to construct a model of choice using preferences, we will examine first the concept by itself.
- Suppose that you want to fully describe the preferences of an agent towards the elements of a given set X .
- What conditions must the description fulfill? What conditions must the description include?

- Preferences should fully specify the attitude of the agent towards each pair of elements in X .
- For each pair, it should provide an answer to the question “How the agent compares the two alternatives?”
- I will present two equivalent ways to present this question, and formulate the required consistency conditions to make the responses “make sense”, i.e. to say that the responses are preferences.

The Questionnaire Q

- Let us think about preferences on a set X as **answers** to a long Questionnaire Q .
- Questions in Q consists of all questions of the type:

$Q(x, y)$ distinct x, y : How do you compare x and y ? Tick one and only one of the following three options:

- ☐ I prefer x to y (denoted $x \succ y$)
- ☐ I prefer y to x (denoted $y \succ x$)
- ☐ I am indifferent (denoted $x \sim y$)

- A legal answer to Q is a response in which **exactly only one of the boxes is ticked** in each question.

Legal Answers (and illegal answers)

- By only allowing one of the three options above we exclude situations like:
 - ☐ They are incomparable.
 - ☐ I don't know what x is.
 - ☐ I have no opinion.
 - ☐ I prefer both x over y and y over x .
- And also dependence on other factors, like:
 - ☐ It depends on what my friends think.
 - ☐ It depends on the circumstances (sometimes I prefer x but sometimes I prefer y)
- And also we preclude intensity of preferences
 - ☐ I somewhat prefer x .
 - ☐ I love x and I hate y .

Consistency restrictions

- Not all legal answers to Q qualify as preferences over X .
- First we consider that the answer to $Q(x, y)$ must be identical to $Q(y, x)$.
- This is what is called a “framing effect”, in which people who are asked to compare two alternatives tend to prefer the first one. We exclude this possibility.
- We also require that the answer to $Q(x, y)$ and $Q(y, z)$ are consistent with the answer to $Q(x, z)$ in the following way: If x is preferred to y and y is preferred to z , then it must be the case that x is preferred to z . If the answers to $Q(x, y)$ and $Q(y, z)$ are indifference, then so is the answer to $Q(x, z)$.

Definition

Preferences on X are a function Q that assigns to any pair (x, y) of distinct elements exactly one of the three values $x \succ y$, $y \succ x$ or $x \sim y$ such that for any x, y, z in X , the following holds.

- No order effect $Q(x, y) = Q(y, x)$.
- **Transitivity:** if $x \succ y$, $y \succ z$ then $x \succ z$. If $x \sim y$, $y \sim z$, then $x \sim z$.

A discussion of transitivity

- How would you react if somebody told you that he prefers x to y , y to z but z to x ?
- What would you prefer?
 - “A weekend in Paris at a 4-star hotel with food quality Zagat 17 for \$574”,
 - OR
 - “A weekend in Paris for \$574 with food quality Zagat 17 at a 4-star hotel”.
- What if I ask you to compare each one of them to
 - “A weekend in Rome at a 5-star hotel with food quality Zagat 18 for \$612”

Systematic Violations of Transitivity

- Even though transitivity is an appealing concept, we are excluding individuals who base their judgements in procedures that cause a systematic violation of transitivity. Even though such procedures make sense.
- **Aggregation as a source of intransitivity:** An individual's attitude might be derived from the aggregation of more basic considerations.
- “Choosing an apartment” based on price, location and size.
- Suppose you have $X = \{A, B, C\}$ and you have that $A \succ_p B \succ_p C$, $B \succ_l C \succ_l A$, and $C \succ_s A \succ_s B$. You form your preferences between two apartments by pairwise majority.
- You will get A is preferred over B , B over C and C over A !

Systematic Violations of Transitivity

- **Similarities:** An individual might express indifference in a comparison between two elements that are too close to be distinguishable.
- Consider “I prefer Black over Light Gray”



- I perform a pairwise comparison of colors, going from Black, decreasing color saturation, until reaching Gray.
- Suppose I can't distinguish between two very close shades.
- $Black \succ LightGray$ but
 $Black \sim Gray1 \sim Gray2 \sim Gray3 \dots Gray100 \sim LightGray$.

Another way to think about preferences

- A second way to think about preferences is through an imaginary questionnaire R consisting of the following type of question:

$R(x, y)$ for all $x, y \in X$, not necessarily distinct. Is x at least as preferred as y ? Tick only one option

☐ Yes

☐ No

- A legal answer is
 - The answer to at least one of the questions $R(x, y)$ or $R(y, x)$ must be yes.
 - In particular, the silly question $R(x, x)$ must be yes.
 - For every x, y, z if the answer to $R(x, y)$ is yes and the answer to $R(y, z)$ is also yes then the answer to $R(x, z)$ should be yes.
- Whenever $R(x, y)$ has a “Yes” answer, we say $x \succsim y$ (x at least as good as y)

The traditional definition of preferences

Definition

Preferences on a set X is a relation $\succsim$ on X (a way to relate elements of X) satisfying:

- **Completeness:** For any $x, y \in X$ either $x \succsim y$ or $y \succsim x$.
- **Transitivity:** For any $x, y, z \in X$ such that $x \succsim y, y \succsim z$ then $x \succsim z$.

Proposition

Both definitions of preferences, either using the questionnaire Q or the questionnaire R are equivalent. A legal answer in Q constitutes a legal answer in R and the consistency requirements for answers in Q are equivalent to the consistency requirements for R

We say that	A response to: $Q(x, y)$ and $Q(y, x)$	A response to: $R(x, y)$ and $R(y, x)$
x is strictly preferred to y	$x \succ y$	Yes and No ($x \succsim y, y \not\succsim x$)
x is indifferent to y	$x \sim y$	Yes and Yes ($x \succsim y, y \succsim x$)
y is strictly preferred to x	$y \succ x$	No and Yes ($x \not\succsim y, y \succsim x$)

Utility functions and the utility representation

- Working with preferences is a bit cumbersome.
- In particular we would like to be able to apply some techniques from calculus for when we model consumer choice.
- So now we look at the questions of when can preferences be represented by a suitable index, and how this index would look like.
- In order to do this, we will work with the concept of utility.

Definition

A **utility function (index)** $u : X \rightarrow \mathbb{R}$ represents a preference relation $\succsim$ over X if and only if, for all $x, y \in X$ we have that $x \succsim y \Leftrightarrow u(x) \geq u(y)$.

- Is it always possible to find such a function? **Yes as long as $\succsim$ is well-behaved.**

Definition

Preferences on a set X is a relation $\succsim$ on X (a way to relate elements of X) satisfying:

- **Completeness:** For any $x, y \in X$ either $x \succsim y$ or $y \succsim x$.
 - **Transitivity:** For any $x, y, z \in X$ such that $x \succsim y, y \succsim z$ then $x \succsim z$.
-
- When a relation $\succsim$ satisfy completeness and transitivity, we call it a **rational preference relation**.
 - This term is misleading, so I prefer to say that a preference that satisfy completeness and transitivity is internally consistent.

When can we find such u ?: X is a finite set

Proposition

Suppose X is finite. Then the assumptions of completeness and transitivity imply that there exist a utility function u that represents the preference relation $\succsim$ on X

- Define for any $x \in X$ $R(x) = \{y \in X : x \succsim y\}$, i.e. those y for which the answer to $R(x, y)$ is Yes.
- We put $u(x) = \text{Number of elements of } R(x)$.

What happens when X is infinite

- We need a technical definition to handle this case:

Definition

Let x_n and y_n be two sequences of elements of X . If $x_n \succsim y_n$ for every n and $x_n \rightarrow x$ and $y_n \rightarrow y$, then $x \succsim y$

- So we have in general:

Proposition

Every continuous preference relation (Rational) can be represented by a continuous utility function.

Utility is a purely ordinal concept!

- The actual number does not matter, only the order it represents.
- If u represents preferences, then for any increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(u(x))$ also represents the same preferences.
- An increasing function is called **monotone**. $f : \mathbb{R} \rightarrow \mathbb{R}$ is monotonic or increasing if and only if for every $x, y \in \mathbb{R}$ such that $x \geq y$, implies $f(x) \geq f(y)$.
- So what this result is saying is that any index that respects the order on X induced by the preference $\succsim$ works as a utility function.
- Rephrasing, if u represents $\succsim$, so does any monotonic transformation of it.

Representing Preferences: Indifference curves

- From now on, we will assume that our set $X = \mathbb{R}^n$, which we will interpret as the set of **consumption bundles**.
- An indifference curve is the set of points that are indifferent between each other:

$$I(x) = \{z \in X | z \sim x\}$$

- If preferences are represented by a utility function, we can express the same idea as follows:

$$I(x) = \{z \in X | u(z) = u(x)\}$$

- A graphical representation of an individual's preferences is by drawing their indifference curve map.

Indifference curve map

Each indifference curve represents a different utility level.

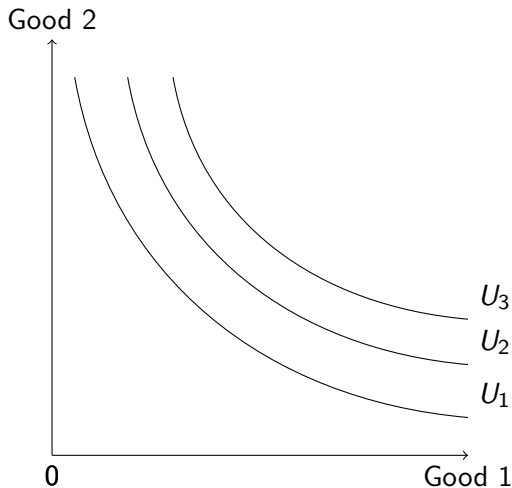


Figure: An indifference map

The set $R(x)$

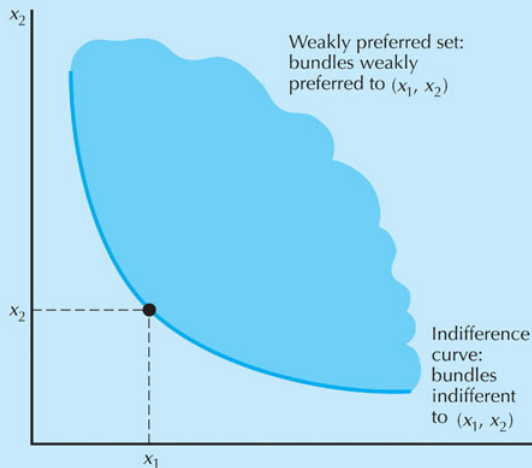
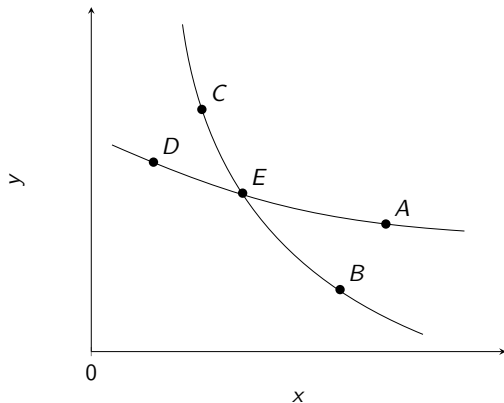


Figure
3.1

Transitivity and indifference curves

If preferences are transitive, indifference curves do not cross!



Suppose $A \succ B$ and $C \succ D$. From the picture, $A \sim D \sim E$ and $B \sim C \sim E$. Then by transitivity $A \sim B$ and $C \sim D$, which is a contradiction.

Homework: Try assuming now that $B \succ A$ and $D \succ C$.

Examples: Perfect Substitutes

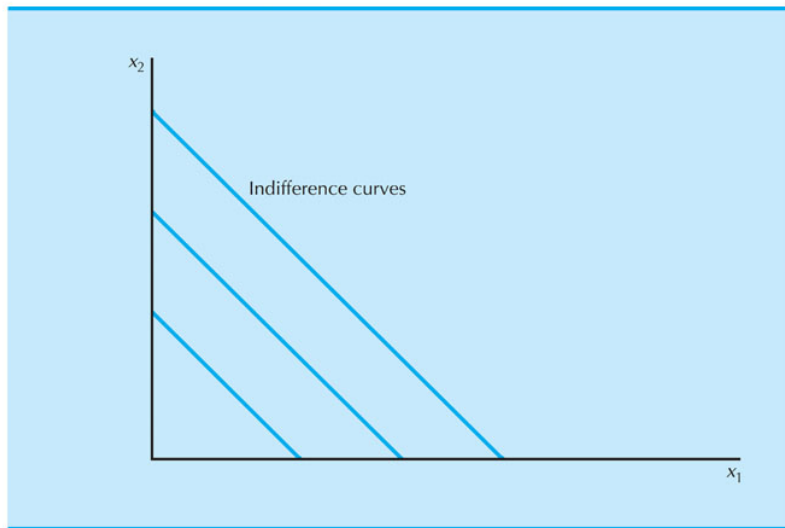


Figure
3.3

Examples: Perfect Complements

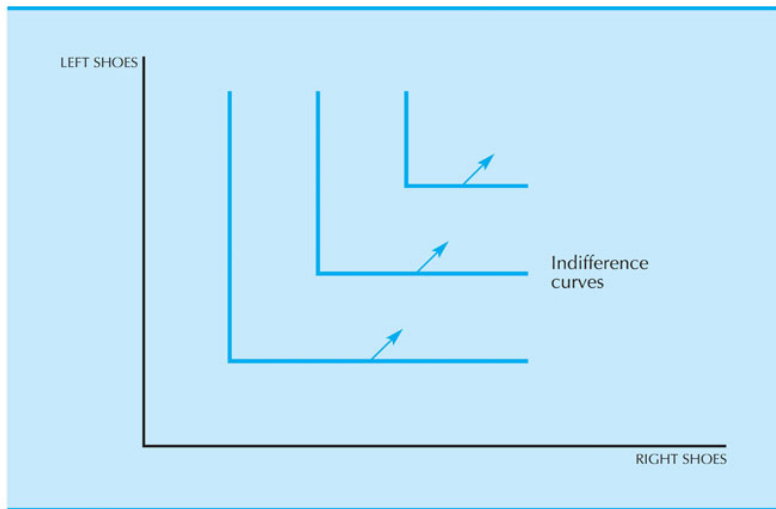


Figure
3.4

Examples: one good and one “bad”

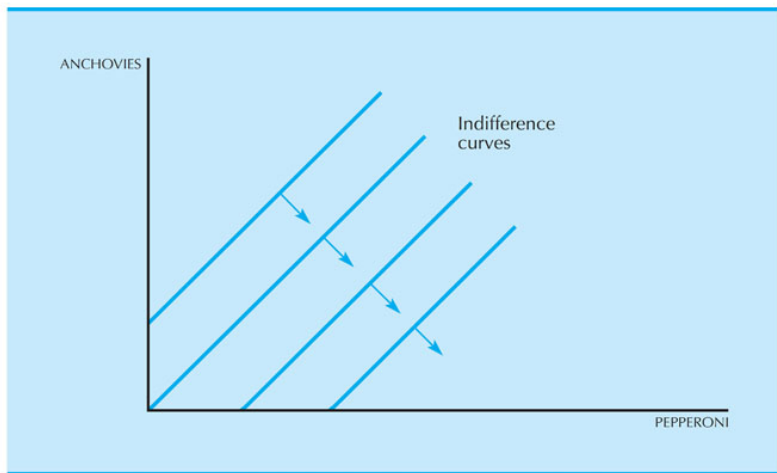


Figure
3.5

Examples: a neutral good

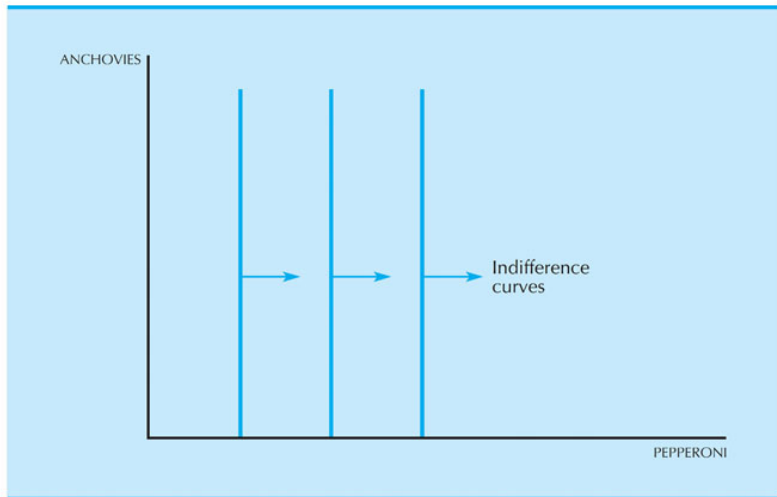


Figure
3.6

Examples: Satiation

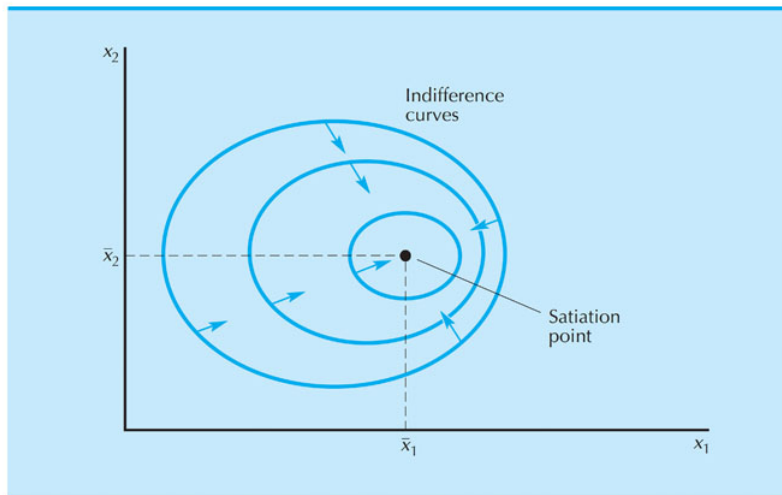


Figure
3.7

Properties of preferences

Definition

Preferences $\succsim$ over $X = \mathbb{R}^n$ are said to be **monotone** (resp. **strictly monotone**) if $x \geq y$ implies $x \succsim y$ (resp. $x \geq y, x \neq y \Rightarrow x \succ y$.)

$x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ are vectors in $\mathbb{R}^n$. We say that $x \geq y$ if and only if the inequality holds component-wise, i.e. $x_i \geq y_i$ for all i . $x \neq y$ means $x_i \neq y_i$ for at least one i .

Definition

A preference relation is convex if for all x and y with $x \sim y$ and for all $\alpha \in [0, 1]$ we have that $\alpha x + (1 - \alpha)y \succsim x \sim y$. **Convexity** means that the at least as good set is a **convex set**

All of these properties can be stated in terms of utility functions.

Monotonic preferences

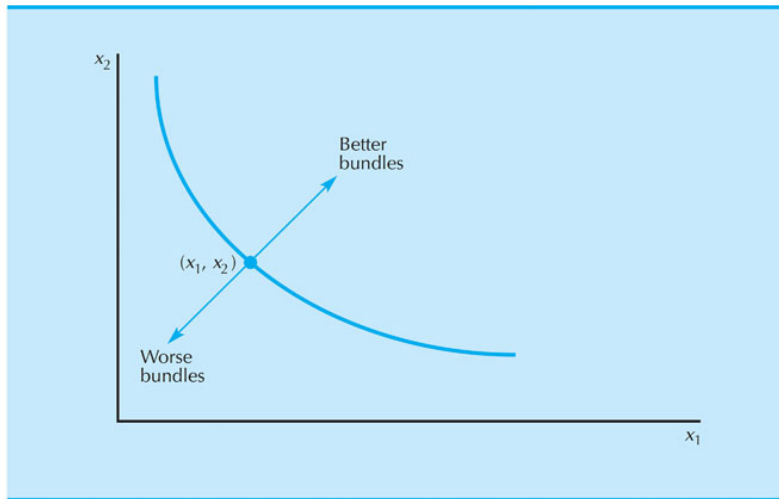
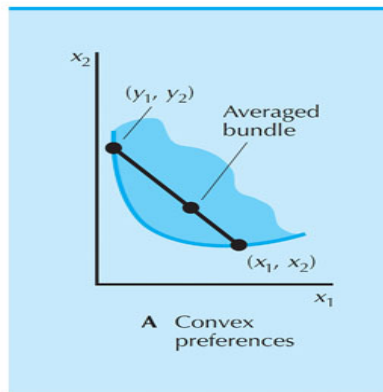


Figure
3.9

Convex preferences



**Figure
3.10A**

Non-Convex preferences

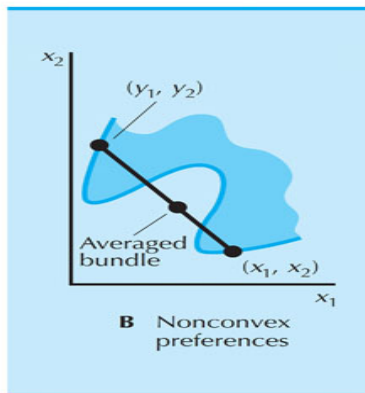


Figure 3.10B

Concave preferences

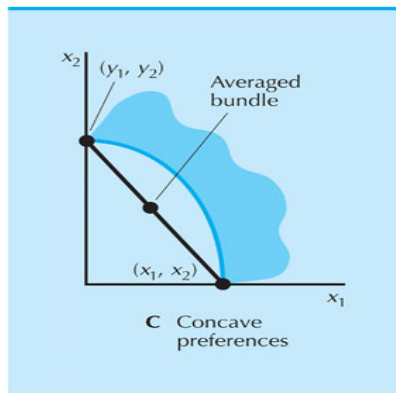


Figure
3.10C

Convexity and balance in consumption

- Intuitively, convexity has to do with an individual preferring more balanced consumption bundles than more extreme ones.
- Suppose you have bundles $(x_1, x_2) \sim (y_1, y_2)$. If preferences are convex, then the bundle $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$ is preferred.
- This is usually assumed when we are talking about consumption bundles that are “economic goods”.
- However, convexity might not hold if the consumption bundle includes an item that might be considered a “bad”, like pollution.

Math detour: Functions of several variables

A multivariate function is identical to the one variable case, but now function has more than one “argument”. In this course, we will only consider functions that map elements from $\mathbb{R}^n$ to $\mathbb{R}$.

Definition

*A multivariate function f is an assignment from some set $A \subset \mathbb{R}^n$ to some set $B \subset \mathbb{R}$ which satisfies the condition that to every element in A , f assigns **one and only one** element from the set B . We write $f : A \rightarrow B$.*

So, a multivariate function defined with respect to $n > 1$ arguments is written as:

$$y = f(x_1, x_2, \dots, x_n)$$

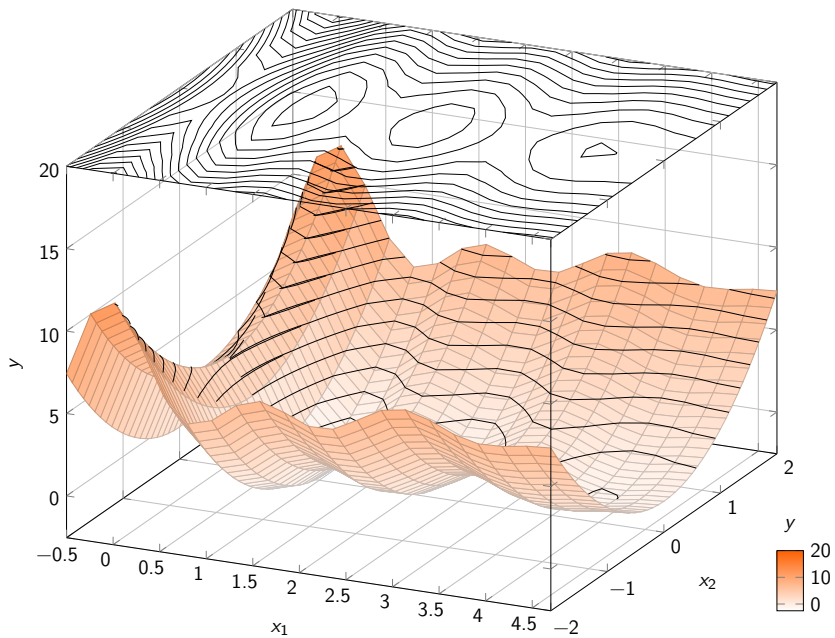
Math detour: Level sets

Definition

Consider the multivariate function $f : A \rightarrow B$ where $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}$. A *level set* of this n -variable real function is a set of the form

$$L_c(f) = \{(x_1, \dots, x_n) \in A \mid f(x_1, \dots, x_n) = c\} \quad (1)$$

where c is a constant such that $c \in B$. This means that it is the set of all elements of the domain such that the function takes on the same value on them.



Math detour: Utility functions are multivariate functions

Consider the function

$$U(x_1, x_2) = \sqrt{x_1 x_2}.$$

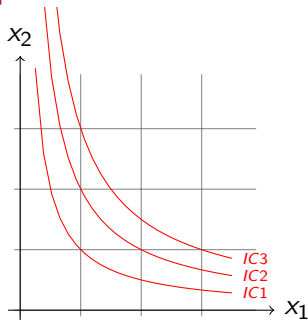
This function gives the level of utility of consuming a consumption bundle composed of x_1 units of good 1 and x_2 units of good 2.

For a given utility level $\bar{U}$ the level sets for this constant satisfy the equation

$$\bar{U} = \sqrt{x_1 x_2}.$$

We call these level sets **indifference curves**, that is, the set of points in which the consumer is indifferent between different bundles.

For different values of $\bar{U}$ we get different indifference curves, which gives rise to the **indifference curve map**.



Math detour: Partial Derivatives

Definition

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a multivariate real-valued function, $f(x_1, x_2, \dots, x_i, \dots, x_n)$. A **partial derivative** of f with respect to x_i at point $\hat{x} = (\hat{x}_1, \hat{x}_2, \dots, \hat{x}_i, \dots, \hat{x}_n)$ is

$$\left. \frac{\partial f(x)}{\partial x_i} \right|_{x=\hat{x}} = \lim_{h \rightarrow 0} \frac{f(\hat{x}_1, \hat{x}_2, \dots, \hat{x}_i + h, \dots, \hat{x}_n) - f(\hat{x}_1, \hat{x}_2, \dots, \hat{x}_i, \dots, \hat{x}_n)}{h} \quad (2)$$

- The partial derivative with respect to x_i can be denoted by $\frac{\partial f}{\partial x_i}$ or f_i also.
- This definition says that when you take the partial derivative with respect to x_i , the other variables are treated as *constants*, that is, you should keep all other variables fixed.
- Thus, there is nothing new here with respect to single variable calculus...

Indifference curves and the marginal rate of substitution

- Consider the case $X = \mathbb{R}^2$. and assume that preferences are represented by u .
- Indifference curve U_1 is $\{(x, y) | u(x, y) = U_1\}$
- The equation $U_1 = u(x, y)$ defines, *implicitly* y as a function of x .
- If u is differentiable, we can think about calculating what is the slope of the indifference curve.
- This slope has a very important economic meaning:

Definition

The **marginal rate of substitution** is the *negative* of slope of an indifference curve U_1 at some point. That is

$$MRS = -\frac{dy}{dx} \Big|_{u=U_1} \quad (3)$$

the MRS tells us about the trades that an individual is willing to make, it is the opportunity cost in terms of y of consuming one unit more of x

The MRS

- We keep assuming that u is differentiable.
- Take the equation for the indifference curve: $U_1 = u(x, y)$
- We totally differentiate both sides with respect to x and y
- $dU_1 = u_x dx + u_y dy$ is the total derivative.
- U_1 is a constant so $dU_1 = 0$.
- Thus $0 = u_x dx + u_y dy$ which implies that $\frac{dy}{dx} = -\frac{u_x}{u_y}$.
- Therefore, $MRS = \frac{u_x}{u_y}$.
- We can also rephrase convexity of preference in terms of the MRS

Definition

Preferences are convex if the utility function that represents them has decreasing MRS.

More on convexity

- Exactly, what kind of preferences satisfy convexity?
- Previously, we assumed that u is differentiable to present the condition for the MRS.
- But in general, what guarantees that u represents convex preferences?
- Notice that rephrased in terms of utility, convexity means.

Definition

A preference relation (represented by u) is convex if for all x and y with $u(x) = u(y)$ and for all $\alpha \in [0, 1]$ we have that $u(\alpha x + (1 - \alpha)y) \geq u(x) = u(y)$.

More definitions

Definition

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is concave if for all $x, y \in \mathbb{R}$ and all $\alpha \in [0, 1]$ we have that $f(\alpha x + (1 - \alpha)y) \geq \alpha f(x) + (1 - \alpha)f(y)$

Definition

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasi-concave if for all $x, y \in \mathbb{R}$ and all $\alpha \in [0, 1]$ we have that $f(\alpha x + (1 - \alpha)y) \geq \min f(x), f(y)$.

- Note that quasi-concavity gives precisely the definition of **convex preferences**.
- Concavity implies quasi-concavity, but not the other way around.
- In economics, we are thus interested in quasi-concave utility functions, because they represent convex preferences.

Utility functions for specific preferences

- Cobb Douglas $u(x_1, x_2) = x_1^\alpha x_2^\beta$.
- Perfect Substitutes $u(x_1, x_2) = \alpha x_1 + \beta x_2$
- Perfect complements $u(x_1, x_2) = \min(\alpha x_1, \beta x_2)$
- CES utility: $u(x_1, x_2) = (\alpha x_1^\delta + \beta x_2^\delta)^{\frac{1}{\delta}}$,
- Quasilinear utility $u(x_1, x_2) = v(x_1) + x_2$

Optimization

- In economics we are usually interested in maximization or minimization problem. Therefore, it is important to be well-versed in optimization techniques.
- The problems that you will encounter in this course are very well-behaved, so we can actually use simple calculus techniques in order to solve them.

Definition

The function f has a **local maximum** at x^* if there exists an interval I such that it contains x^* and $f(x^*) \geq f(x)$ for all $x \in I$.

Definition

The function f has a **global maximum** at x^* if $f(x^*) \geq f(x)$ for all x in the domain of f .

First and Second order conditions

Theorem (First Order Necessary Condition for maximum/minimum)

Let f be a real-valued function of one variable. Suppose that f has a local maximum/minimum at some point x^ and suppose that $f'(x)$ exists. Then $f'(x^*) = 0$.*

Theorem (Second Order Sufficient condition for a maximum(minimum))

Let f be a real-valued function of one variable. Suppose that $f''(x^)$ exists and suppose that $f'(x^*) = 0$ and $f''(x^*) < 0$ ($f''(x^*) > 0$). Then f has a local maximum(minimum) at x^* .*

Multivariate function maximization

Theorem (First Order Necessary Condition for maximum/minimum)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Suppose that f has a local maximum/minimum at some point x^ and suppose that f is differentiable. Then $\frac{\partial f(x_1, \dots, x_n)}{\partial x_i} = 0$ for all $i = 1, \dots, n$*